Hugging Face API Data Extraction Methodology

1. Purpose and Scope

The objective of the Hugging Face data extraction was to collect grey literature relating to artificial intelligence models, with particular emphasis on security-relevant capabilities, risks, and mitigation mechanisms. The Hugging Face platform was selected due to its central role within the open AI ecosystem, functioning as a primary distribution hub for pre-trained models used across both research and production environments.

In contrast to GitHub, which predominantly hosts source code repositories, Hugging Face provides direct access to model artefacts, associated documentation, and usage metadata. This enables the analysis of deployed and reusable AI systems rather than development infrastructure alone. Accordingly, Hugging Face was treated as a complementary grey literature source, enriching the broader dataset with artefact-level insights into AI capabilities.

2. Data Source

Data were collected using the Hugging Face Hub API, which exposes metadata for publicly available models hosted on the platform. The extraction focused exclusively on public model repositories, ensuring transparency and alignment with the principles of open research.

The API responses provided structured metadata, including download numbers, repository URLs, tags, and license classifications where available.

3. Search Strategy and Keyword Selection

3.1 Conceptual Focus

The search strategy was designed to identify models relevant to:

- Natural language processing
- Representation learning and embeddings
- Large language models
- Sentiment analysis and classification
- Multilingual and speech-related AI systems

These areas were selected due to their relevance to AI capability development, downstream deployment risk, and potential security or misuse considerations.

3.2 Keyword-Based API Queries

The Hugging Face API does not support complex Boolean search logic, which is comparable to that found in academic databases. Instead, a keyword-based querying approach was employed, using targeted search terms aligned with the conceptual focus areas above.

Keywords were selected to prioritise high-impact and widely reused model categories, rather than narrow or experimental tasks. Queries were executed programmatically, and results were aggregated into a unified dataset.

4. Data Extraction Process

The extraction was implemented via a scripted API pipeline that:

- Submitted keyword-based queries to the Hugging Face Hub API
- Retrieved metadata for matching model repositories
- Normalised responses into tabular format for downstream analysis

At the conclusion of this process, the dataset consisted of 218 model records.

No enrichment, filtering, or relevance scoring was applied during this stage. The dataset, therefore, represents the raw output of the API search, subject only to the constraints of the Hugging Face API itself.

5. Secondary Filtering and Classification

To refine the dataset towards security-relevant models, a second-stage filtering process was implemented using a Python-based classification pipeline. This stage introduced a keyword-driven relevance assessment based on the presence of security-related terminology within extracted metadata.

5.1 Keyword Taxonomy

Two categories of keywords were defined:

- **Strong keywords (direct security relevance):** security, vulnerability, vulnerabilities, adversarial, attack, attacks, poison, poisoning, safety
- **Weak keywords (indirect or contextual relevance):** evaluation, limitations, robust, risk, risks, mitigation

5.2 Classification Logic

Each model record contained a list of detected keywords. These were evaluated using a rule-based classification function:

- Models containing at least one **strong keyword** were classified as *strong matches*
- Models containing only weak or generic terms were classified as *weak matches*

This approach prioritised precision over recall, ensuring that retained models demonstrated explicit and direct relevance to AI security, adversarial robustness, or safety mechanisms.

5.3 Filtering Outcome

Following classification:

- The dataset was partitioned into strong and weak subsets
- Only strong matches were retained for further analysis

This process reduced the dataset from **218 models to 15 security-relevant models**, representing a highly focused subset aligned with the research objective.

To support transparency and reproducibility:

- The full classified dataset was preserved
- A filtered dataset containing only strong matches was exported separately

6. Methodological Considerations and Limitations

Several methodological constraints were identified:

- Hugging Face metadata quality varies significantly across repositories
- Not all models provide detailed documentation or clear usage intent
- API search functionality is less expressive than GitHub's repository search

These limitations underscore the need for manual inspection and expert judgment in later stages of analysis.

7. Summary

The Hugging Face API extraction yielded a structured but unfiltered dataset of 218 publicly hosted AI models, serving as a valuable complement to GitHub repository data. A secondary classification stage refined this dataset to 15 models demonstrating explicit security relevance. By deferring filtering and validation, the methodology preserves transparency and reproducibility while enabling informed, expert-led refinement in subsequent phases of the project.